Paleomagnetism of Carboniferous strata from the Stellarton Gap

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Abstract: Deposition of Carboniferous sediments within the Stellarton Graben of Nova Scotia was controlled by displacements on the bounding Hollow and Cobequid fault systems. Periods of both normal and transcurrent faulting have occurred on both faults. As a result, models for the development of the Stellarton Basin have included a pull-apart basin and a thermal subsidence basin. Paleomagnetic study of stratigraphic sequences from within the graben showed evidence for three separate periods of remanence acquisition (ST3 – ST2 – ST1). Ordering the acquisition sequence for these three magnetizations is mainly circumstantial, being based on the distribution of each remanence phase relative to stratigraphic position and deformation of the strata. The oldest magnetization ST3 is grossly divergent from previously reported Carboniferous results suggesting an early period of large-scale clockwise rotation. Magnetization ST2 has a similar pole position to many others from Maritime Carboniferous basins located within the Appalachian orogenic belt. The divergence of these poles from the cratonic poles is interpreted as indicating that these magnetizations were acquired at a time when the reference horizontal was different to the present horizontal. The third magnetization ST3 is identical to a large number of late-stage Carboniferous regional overprint magnetizations. In summary, the paleomagnetic evidence suggests that the Stellarton Basin was formed in an early (Westphalian B to C) pull-apart phase that was followed by a later thermal subsidence phase.

Résumé : Durant le Carbonifère, la déposition de sédiments à l'intérieur du graben de Stellarton, en Nouvelle-Écosse, était contrôlée par des déplacements sur les systèmes de failles de Hollow et de Cobequid. Des épisodes de failles normales et de failles décrochantes ont eu lieu sur les deux failles. Les modèles pour le développement du bassin de Stellarton comprennent ainsi un bassin d'extension et un bassin de subsidence thermale. Une étude paléomagnétique des séquences stratigraphiques à l'intérieur du graben montrent des évidences pour trois périodes distinctes d'acquisition de rémanence (ST3 – ST2 – ST1). L'ordonnancement de la séquence d'acquisition pour ces trois magnétisations est surtout circonstanciel, étant donné qu'il est basé sur la distribution de chaque phase de rémanence par rapport à la position stratigraphique et la déformation des strates. La magnétisation la plus ancienne ST3 est très différente des résultats antérieurement rapportés pour le Carbonifère, suggérant une période précoce de rotation, à grande échelle, dans le sens horaire. La position du pôle de la magnétisation ST2 est semblable à celle de bien d'autres dans des bassins carbonifères des Maritimes situés à l'intérieur de la ceinture orogénique appalachienne. La divergence entre ces pôles et les pôles cratoniques indiquerait que ces magnétisations ont été acquises à une période où l'horizon repère différait du présent horizon. La troisième magnétisation ST3 est identique à un grand nombre de magnétisations de surimpression régionales qui ont eu lieu lors d'étages tardifs du Carbonifère. En résumé, les évidences paléomagnétiques démontrent que le bassin de Stellarton aurait été formé dans une phase d'extension précoce (Westphalien B à C), laquelle aurait été suivie d'une phase de subsidence thermale postérieure.

[Traduit par la Rédaction]

Introduction

Genesis of the upper Paleozoic Maritimes Basin (Fig. 1) of eastern Canada has been explained by two tectonic models. Belt (1968) proposed a simple rift model, while Bradley (1982) suggested the Maritimes Basin was associated with dextral strike-slip faulting that produced a pull-apart basin. Evidence has been advanced in favour of both options. McCutcheon and Robinson (1987) explain the development of the Maritimes Basin as being purely of passive subsidence origin, while Gibling et al. (1987) support a strike-slip fault model. The conflicting views arise from different interpretations on the timing and magnitude of fault displacements and the effects these motions had on sedimentation. Boehner et al. (1986), followed by Ryan et al. (1987), offered a compromise position in which the conflicting views can be harmonized by resolving the development of the Maritimes Basin into two distinct phases. Calder (1998) summarized the importance of a "mid-Carboniferous event" as a pivotal change between two stages of sedimentary basin evolution. The first stage, extending from the Upper Devonian to mid-Pennsylvanian, is considered to be the pull-apart phase and is related to regional tectonic transpression (Bradley 1982; Keppie 1982). The second phase of basin development, from late Pennsylvanian to Early Permian, during which strata overlapped from the

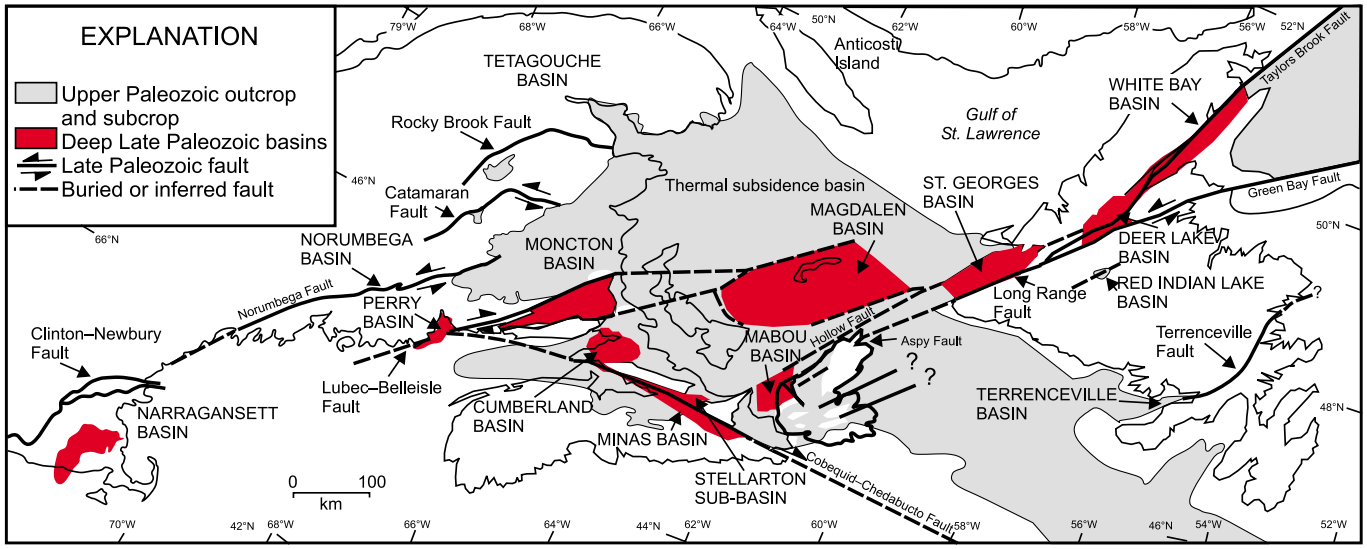
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Fig. 1. Distribution of Carboniferous sedimentary basins in the Maritime region of Canada and their relationship to regional fault structures. Regions of thermal subsidence are differentiated from deeper basins directly associated with fault structures. Diagram modified from Ryan et al. (1987).



earlier basins onto the surrounding basement, is considered as being a regional thermal subsidence phase. Hamblin and Rust (1989), Hamblin (1992), Martel and Gibling (1996), and Pascucci et al. (2000) present reports on the sedimentological evolution of differing Carboniferous sequences within the Maritimes. In all of these examples, initial sedimentation is attributed to the development of half-grabens in a transcurrent tectonic setting, followed by a period of basin development associated with a regional thermally driven sag.

The two models can be differentiated on the basis of their tectonic effect on the basin infill (Webb 1969). Progressive development of a pull-apart basin produces incrementally increasing rotations of sequentially older strata deposited in the evolving basin (Crowell 1974; Allen and Allen 1990, p.117; McKenzie and Jackson 1986). In contrast, continuing development of a thermal subsidence-generated basin involves no systematic rotation of the basin sediments. In this paper, we test the applicability of these models by paleomagnetically examining samples from stratigraphically controlled sections in the Stellarton Sub-Basin, which lies at the juncture of the east–west-trending Minas Geofracture and the NE–SW-trending Hollow Fault (Figs. 1, 2).

Bell (1940) considered the faults bounding the Stellarton Graben to be post-depositional normal faults. Following more detailed mapping of structures within the fault zone, Eisbacher (1967, 1969) demonstrated there had been significant dextral movement on the western Cobequid Fault during Pennsylvanian. More recently, using integrated structural and stratigraphic mapping, Yeo and Gao (1986, 1987) have identified "three phases of basin subsidence and filling" and "suggest three syn-sedimentary pulses of faulting" within the period Westphalian C to D. Most recent mapping of the Carboniferous stratigraphy in the Stellarton Gap has been reported by Chandler et al. (1997).

Geological setting

Yeo (1985) reviewed the regional geology of Stellarton Gap and the adjacent highlands. The Cumberland Group of Westphalian B age includes two formations in the Stellarton Gap: the New Glasgow conglomerate and the Middle River Formation. In the area examined in this survey, the Cumberland Group rests unconformably on the underlying Canso Group. The uppermost Carboniferous unit of the Stellarton Gap is the Pictou Group, which is characterized by gray, medium- to coarse-grained sandstone with interbedded red mudstone. Spore assemblages from coals within the Pictou Group indicate the age of this unit ranges from late Westphalian C to Stephanian A.

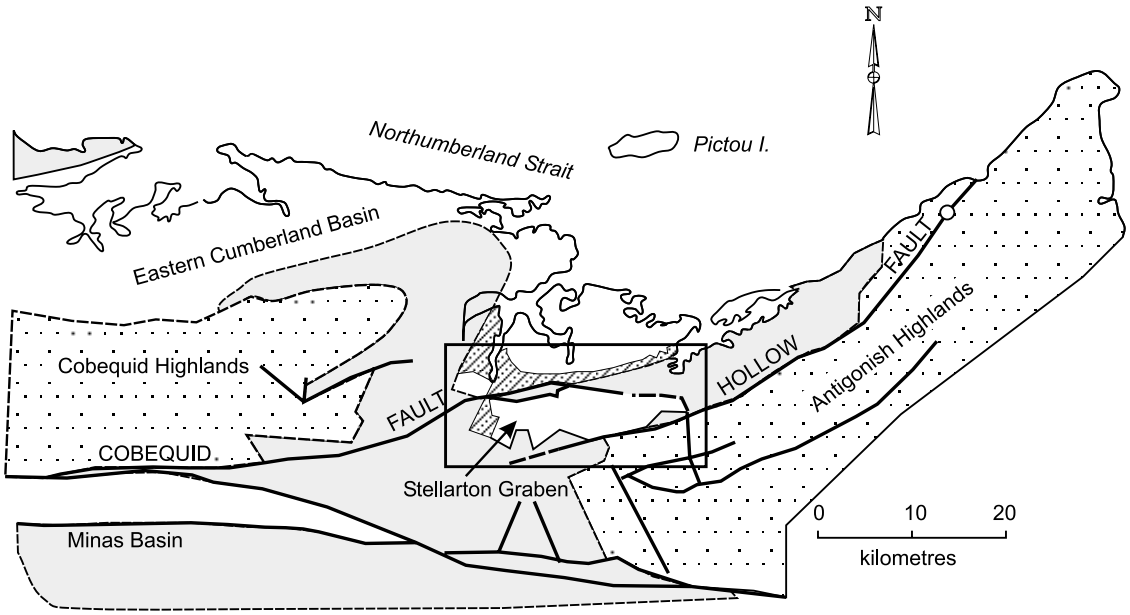
Confined to the Stellarton Graben are a thick sequence of coal and oil-bearing shale and lithic sandstones that are known as the Stellarton Group (Fig. 2*a*). Bell (1940) recognized six members in the Stellarton Group. In ascending stratigraphic order, they are the Skinner, Westville, Plymouth, Albion, Coal Brook and Thorburn members. Spore assemblages from Stellarton Group coals range in age from Westphalian C to Westphalian D. This group then is partly time-correlative with the Pictou Group. Locally, the Stellarton Group rests disconformably or conformably on the Cumberland Group, or unconformably on the Canso Group.

Sample collection and treatment

Samples were collected along four stream sections, each of which covered a particular aspect of the Stellarton Basin geology (Fig. 2*b*). The two most westerly sections along Middle River and Skinner Brook covered the stratigraphic relationship between the underlying Cumberland Group and the overlying Skinner member of the Stellarton Group. Bell (1940) reported a disconformable relationship between these two units. The next easterly section along the East River examined the relationship between the underlying Canso Group strata and the overlying Stellarton Group, which locally contains a well-defined Plymouth member. The unconformable relationship between the Canso Group and the Stellarton Group is well-exposed along the banks of the East River. Along McLellan's Brook, we examined the

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Fig. 2*a*. Map showing distribution of the Hollow and Cobequid Faults and their location relative to the study area. Based on Yeo and Gao (1987).



relationship between the Windsor Group (Forbes Lake conglomerate), the Canso Group, and the overlying Stellarton Group. Samples from 28 sites were examined. All samples were field-drilled and oriented using a sun compass prior to removal from the outcrop.

All remanence measurements were made on a CTF cryogenic magnetometer at the University of Windsor, Windsor, Ontario. During the period of specimen measurement, the background shaft remanence of the cryogenic never exceeded 1×10^{-5} A/m. All specimens were individually stepwise demagnetized using a minimum of 12 steps per specimen. Thermal treatments were performed using a Schonstedt TSD-1 thermal demagnetizer. Prior to and during the demagnetization process the specimens were stored in the magnetically shielded room of the cryogenic system.

The demagnetization trajectories resulting from the stepwise thermal treatment were visually examined using orthogonal projection plots, and numerically dissected using a version of Kirschvink's (1980) principal component analysis routine. Site mean directions were calculated using standard Fisher (1953) paleomagnetic statistics from specimen vectors having maximum angular deviations (MAD) of less than 10°.

Demagnetization results

Most specimens exhibited multi-component demagnetization trajectories on orthogonal plots: the individual components present in each specimen were directly related to the stratigraphic unit from which it was obtained, the nature of the lithology, and the tectonic history of the sample site.

Windsor Group's Forbes Lake conglomerate

Underlying the Canso Group on McLennan's Brook is a conglomeratic unit, which has been tentatively assigned to the Forbes Lake Formation. The cobbles in the conglomerate at this locality are mainly of volcanic origin and probably were derived from the uppermost Devonian McAras Brook Formation. At this outcrop, 11 cores provided one site for paleomagnetic analysis. After stepwise alternating field (AF) demagnetization up to a maximum field of 110 nT, each specimen was then subjected to 15 steps of thermal demagnetization up to a maximum temperature of 690°C. None of the specimens defined simple linear demagnetization trajectories. To define the directions present in these samples, it was necessary to employ the great circle remagnetization and intersection method (Palmer et al. 1981). This methodology works because the cobbles in the conglomerate have randomized primary remanence directions. Two periods of magnetic overprinting were detected in the cobbles of the Forbes Lake conglomerate. The first component, with a direction of declination (D) = 200°, inclination (I) = –10 (N = 6 specimens) was generally found by AF demagnetization up to ~70 nT. The second component, with an estimated direction of D = 217°, I = +26° (N = 5 specimens) was usually detected by low-temperature thermal demagnetization after AF treatment. None of the specimens retained any meaningful remanence direction after treatment at higher temperatures.

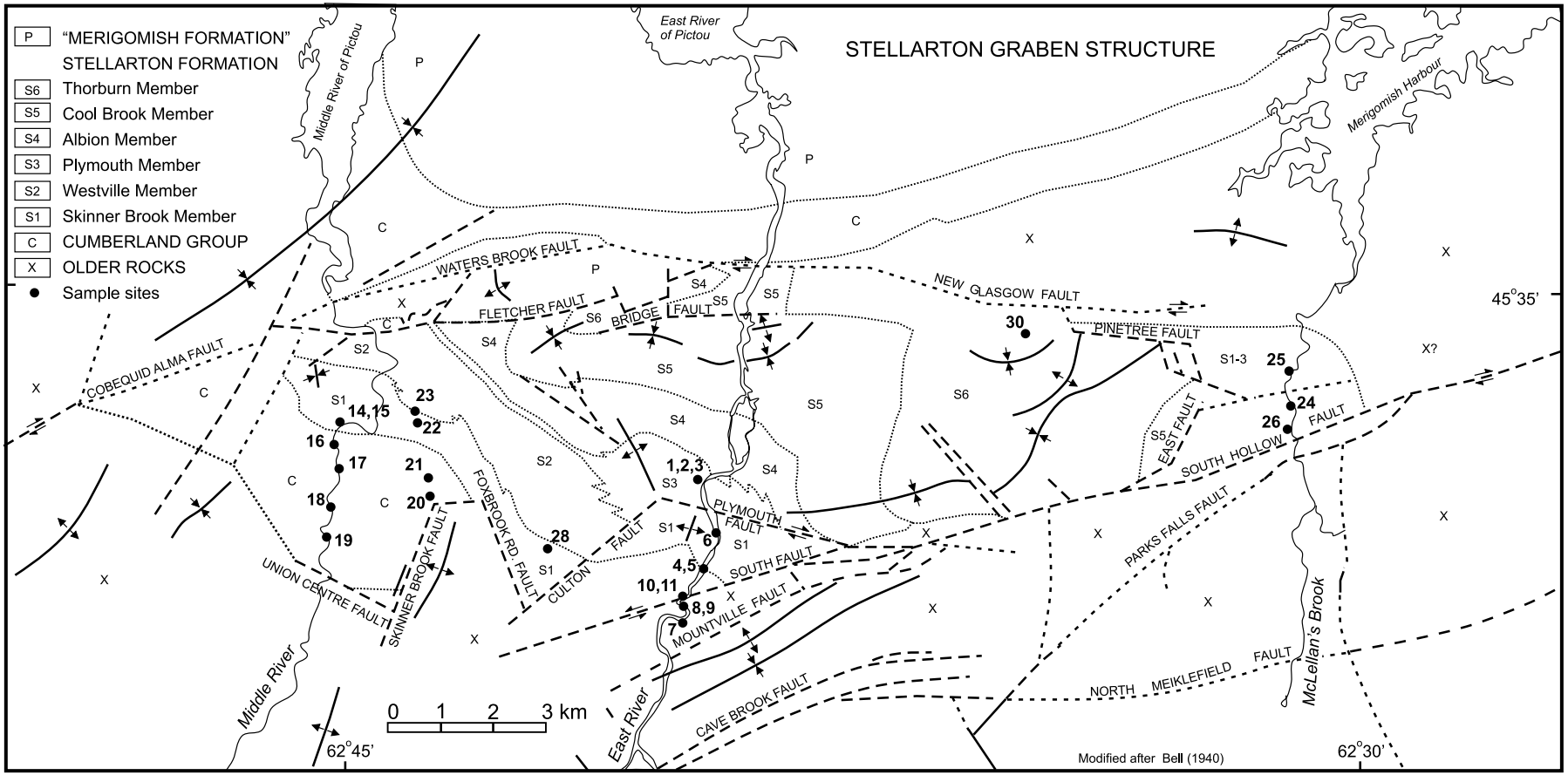
Canso Group

Of the seven sites collected in the Canso Group, six are from a stratigraphic section along the East River (Table 1). Most of the samples were taken from medium-grained brown to red sandstones. Sites 8 and 9 are located in a zone where the Hollow Fault crosses the East River. At this locality, the rocks are highly cleaved, deformed, and intensely hematized (see discussion in Yeo and Gao 1985). With the exception of the lowermost site 7, the remainder of the sites in this section are from rocks dipping to the north.

Away from the Hollow Fault (sites 4, 10, and 11), the NRM remanence is directed easterly with shallow positive to shallow negative inclination. With progressive thermal demagnetization, all specimens migrate towards shallower inclinations (Fig. 3). For the most, part these small vector migrations represent the removal of the steeply positive present

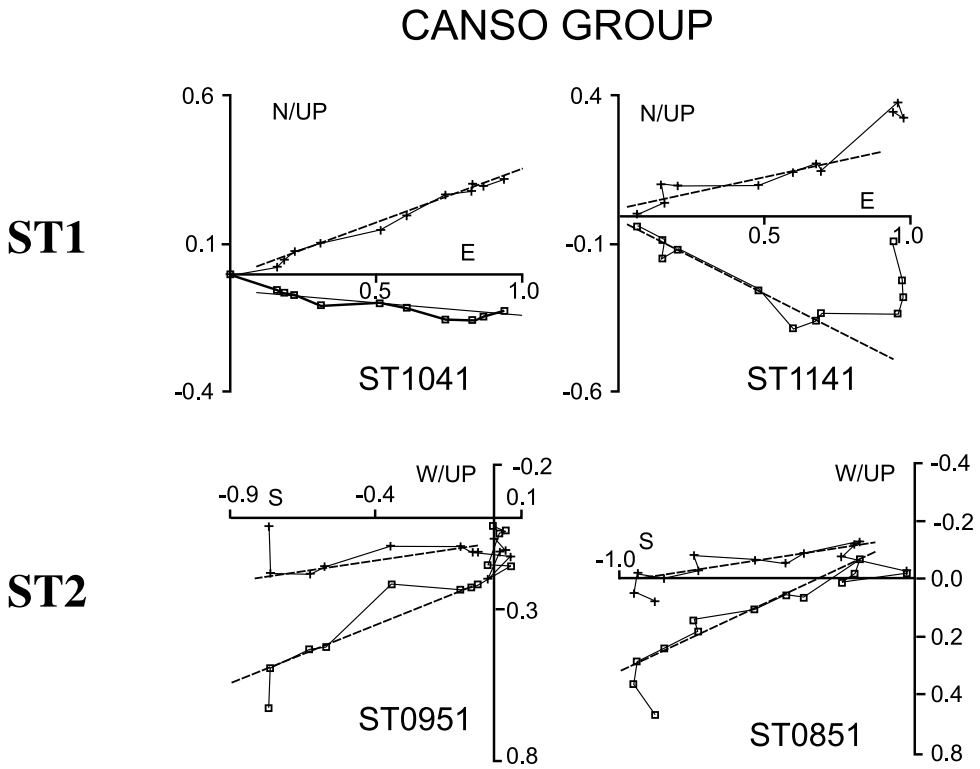
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Fig. 2*b*. Map of Stellarton graben structure showing location of paleomagnetic sampling sites. Stratigraphic units are referenced to stratigraphic column. Local structures are outlined by presence of anticline and syncline axes. (Map based on Bell (1940)).



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Fig. 3. Vector diagrams demonstrating thermal demagnetization characteristics of samples from the Canso Group. Samples ST1041 and ST1141 define the moderately inclined easterly ST1 direction up to Curie Point of hematite. Samples ST0951 and ST0851 have more southerly directed ST2 remanence component. The easterly ST1 component is present in the high-temperature fraction of specimen ST0951.



Earth's field direction. Principal component analysis of the site 11 specimens suggests the presence of two easterly magnetizations: one with a very shallow negative inclination that is similar to the other results included in Table 1; and a second higher temperature component with a site mean in situ direction of $D = 071°$, $I = –27°$ ($N = 4$ specimens, precision parameter (k) = 115, confidence level $\alpha_{95} = 9°$). The significance of this direction will be discussed later in the text.

Towards the Hollow fault zone, a southerly positive inclination vector (Table 1) progressively overwrites the easterly direction. Sites 7 and 9 retain traces of the easterly vector, while at site 8 from the middle of the fault zone all evidence of the easterly direction has been erased (Fig. 3).

Cumberland Group

Of the six sites sampled in the Cumberland Group, only four sites from the Middle River section gave well-defined demagnetization trajectories (Table 1). No meaningful site directions could be obtained from the two sites (20, 21) sampled from the Skinner Brook section. The Cumberland outcrop on Skinner Brook appears to be less reddened and less lithified than the equivalent strata on Middle River.

Samples of the Cumberland Group contain the record of two southerly directed remanence components (Table 1). The two components have average in situ directions of (1) $D = 168°$, $I = +28°$; and (2) $D = 175°$, $I = +01°$. With progressive thermal demagnetization, the vector always migrates towards a shallower positive, and then increasingly negative, inclination (Fig. 4). Some specimens, as in Fig. 4, contain what is probably a low-temperature present Earth's field overprint. When present, the moderate positively inclined vector is always the lowest temperature component, and is always present up to at least 500°C. In the occasional specimen, this vector is retained up to the Curie Point of hematite. More often, it is replaced by the shallow subhorizontal vector, which is then retained to 690°C. When the moderately inclined positive vector is absent, the vector will migrate upon demagnetization from shallow positive inclination to a shallow negative inclination at higher temperatures.

Stellarton Group

Skinner Member

All of the seven sites occupied in the Skinner Member gave interpretable paleomagnetic results (Table 1). The Skinner Member contains the same two components as detected in the underlying Cumberland Group. Moreover these specimens exhibit the same remanence decay characteristics (Fig. 5). The Skinner Member has two features, which distinguish it from the Cumberland Group data. First, site 22 on Skinner Brook has a well-defined normal polarity magnetization, which is oriented exactly anti-parallel to the more common reverse-polarity magnetizations (Fig. 5). As well as being absent in the older Cumberland strata, this normal polarity component is not detected in the stratigraphically equivalent rocks of the Skinner Member on Middle River. Second, the remanence components isolated in the Skinner Member are similar to those in the Cumberland Group. Yet the tectonic dip at sites in the stratigraphically older Cumberland group appear to be systematically *shallower* than the dips recorded

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Table 1.

Site	Bedding	ST3	ST2	ST1
Stellarton Group - Thorburn Member				
	32/130	190, +13 (4,16,24)		
Stellarton Group - Plymouth Member				
3	19/038	187, +08 (3,73,15)		
2	20/038	186, +02 (4,321,5)		
1	18/038	197, –09 (4,120,8)		
Stellarton Group - Skinner and Westville Members				
6	12/065	175, +05 (5,31,14)		
1	12/065	179, +13 (9,93,5)		
12	35/020	169, –05 (6,307,4)	166, +25 (4,167,7)	
13	35/020	163, –02 (6,53,9)	163, +22 (6,36,11)	
14	30/020	174, +11(4,49,13)	171, +34 (7,57,8)	
15	30/020	171, +09(6,58,9)	164, +31 (5,31,14)	
23	32/020	159, –06 (6,90,7)	164, +31 (4,42,14)	
22	36/012	347, +07 (4,478,4)		
25	25/040		165, +32 (3,28,24)	
2	58/008	172, +13 (5,12,23)		
Cumberland Group				
16	32/020	179, –10 (7,128,5)	171, +22 (5,47,11)	
17	26/025		169, +21 (6,38,11)	
18	27/022	172, –16 (5,46,11)	169, +21 (6,79,8)	
19	27/022	163, +01 (6,32,12)	163, +33 (6,20,6)	
Canso Group				
4	53/006			091, +11(3,14,34)
11	48/329			087, –03(5,18,19)
10	38/331			070, –16(3,140,10)
8	25/294		175, +30(5,85,8)	
7	45/147		174, +29(4,28,17)	082, +30(3,282,7)
24	54/010			077, +05(6,49,10)
Windsor Group - Forbes Lake conglomerate				
26	50/022	200, –10 (6 gt. circ)…	217, +26 (5 gt. circ)…	

Note: All directions are quoted relative to the present horizontal. Bedding is quoted as dip/dip direction. Format for the site mean remanence directions is declination, inclination, (# of specimens, precision parameter k, α_{95}). No coherent remanence directions were detected at sites 9, 20, and 21. gt. circ., Great Circle.

in the *younger* Skinner Member sites. Discussion of the possible significance of this difference is deferred to a later section.

Demagnetization of specimens from sites 5 and 6 from gray beds of the Skinner Member, which outcrop on the East River, revealed only the presence of the single shallow positively inclined vector. When heated to about 500°C, all of the specimens exhibited a dramatic increase (>200%) in remanent intensity (Fig. 5). Concurrent with this intensity rise, the remanence direction became randomly oriented. These gray rocks contain no evidence of the more steeply inclined component.

Plymouth Member

All of the three sites occurring in red beds of the Plymouth Member only contained a record of the shallowly inclined remanence component. And as shown in Table 1, the individual sites retained univectorial remanence directions.

Discussion

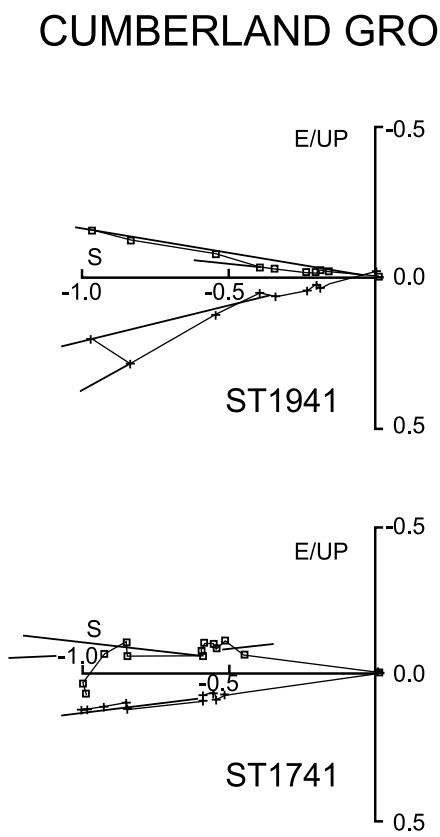
Sequence of remanence acquisition

The Canso Group is the oldest horizon sampled extensively in the Stellarton Basin. Although it is not possible to show that the easterly direction (ST1) dates from the time of Canso deposition, it is possible to show that the magnetization probably predates the subsequent deposition of the Stellarton Group (Table 2). First, complete tilt correction of the in situ site mean directions produced an increase of Fisher precision from 19 to 32 (for $N = 5$ sites). Maximum Fisher (1953) precision (>38) obtained with 80% partial tectonic correction is indicative of an early syntectonic time of acquisition for this remanence (Fig. 6). Second, the easterly oriented direction (ST1) was not detected in any of the younger units examined in this collection. Site 5 in the Stellarton Group is less than 1 m away from site 4 in the underlying Canso Group. There is no trace of the easterly vector at site 5. Therefore, the remagnetization explanation is highly improbable. The angular unconformity with the overlying Stellarton Group indicates that the Canso was folded, and therefore ST1 was acquired, prior to deposition of the Stellarton Group.

This configuration of geological elements, with *a fold* truncated by an erosional surface, where there are different remanence directions above and below the unconformity surface, should appropriately be termed an unconformity test

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Fig. 4. Modified vector diagram demonstrating thermal demagnetization characteristics of specimens from the Cumberland Group. In this figure, the convention placement of the +Y vector has been changed to permit better resolution of the vertical and horizontal components. After removal of a steeply inclined present Earth's field vector the vector decays linearly to the origin.



of remanence acquisition age. Key to the success of the unconformity test is the ability to prove that the magnetization in the underlying unit must predate the unconformity. The unconformity test is only valid if similar rock types are involved with similar porosities, magnetic minerals, etc., above and below the unconformity. If an aquitard was above an aquifer, remagnetization of the aquifer by fluid flow might well postdate deposition of the overlying unit (D.T.A. Symons, personal communication, 2002). For this example, the partial fold test demonstrates that remanence acquisition took place prior to the later sediments being deposited. Hence, acquisition of this easterly directed magnetization must predate deposition of the Stellarton Group. For acquisition of this vector to postdate deposition of the Stellarton requires appealing to some unlikely process by which only the Canso was selectively remagnetized.

The relative age of acquisition of the two remaining remanence directions are not tightly controlled by any geological field tests. Based on circumstantial evidence the probable sequence of acquisition in the Stellarton Graben was ST2 – ST3. The ST2 component was not detected in any of the sites sampled in the youngest (Plymouth and Thorburn) members of the Stellarton Group. The ST3 component was detected in the older units. Therefore, it could be argued that acquisition of ST2 predated deposition of these younger units. The age of the ST2 magnetization relative to deposition of either the Cumberland Group or the Skinner Member of the Stellarton Group is only weakly constrained by a fold test because all of these sites have similar geological attitudes. Complete tilt correction produces a reduction of Fisher precision from 167 to 132 (N = 10 sites). An incremental fold test performed on 10 sites from the Cumberland Group and Stellarton Group – Skinner Member exhibits a broad maximum k value >175 after 10% to 20% unfolding (Fig. 6). Including results from the sites in older rock units confirms that acquisition of this magnetization postdates deformation of the Cumberland Group and Skinner Member. Repeating the incremental fold test with the inclusion of sites 7 and 8, which are from the Canso Group, where it is deformed in the Hollow fault zone, shows that the application of *any* tilt correction produces a significant reduction in Fisher precision (Fig. 6). The ST2 component is also found as a weakly defined overprint in cobbles of the Windsor-age Forbes Lake conglomerate. Circumstantial evidence, therefore, suggests that ST2 was acquired after deformation of the Cumberland Group and the Skinner Member, but before deposition of the younger Plymouth and Thorburn members of the Stellarton Group. One site (22) in this group records a normal polarity magnetization suggesting this magnetization was acquired prior to the onset of the Kiaman reversed-polarity Superchron (DiVenere and Opdyke 1991; Opdyke and Channell 1996; Buchan and Chandler 1999). However, with the limited number of specimens (4) at site 22, the negative fold test on the reversed polarity sites, and the lack of evidence for stratigraphic continuity of the normal polarity horizon this must remain a scientific curiosity (Fig. 7A).

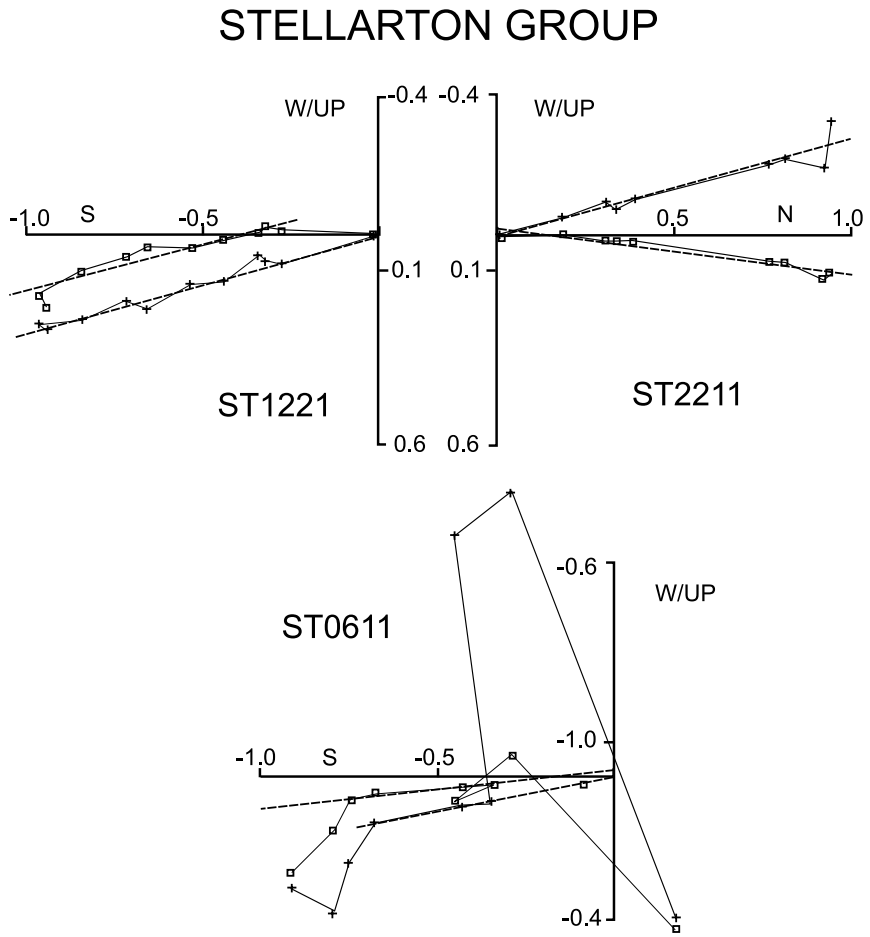
Magnetization ST3 represents the youngest remanence acquisition event recorded in this study (Table 2). It is the only direction found in the youngest units sampled in the Stellarton Basin, the Plymouth and Thorburn members (Table 1). Applying a full-tilt correction to this magnetization, and calculating a mean vector irrespective of whether the site was located in the Cumberland or Stellarton groups leads to a reduction in Fisher precision. An incremental fold test shows a weakly defined maximum after 20% unfolding, but as seen from Table 2, the partially corrected mean direction is not significantly different to the in situ direction. This suggests that ST3 is a secondary post-folding magnetization. The presence of both polarities of this component together with the agreement of the corresponding pole with other published results (Roy and Morris1983) indicate this magnetization is typical of the widespread early Permian Kiaman remagnetization event.

Local tectonic implications

DiVenere and Opdyke (1991), following some earlier work by Roy and Morris (1983), present a convincing case for a systematic progressive change in mean direction with decreasing age during the Carboniferous. Buchan and Chandler (1999), who found a similar paleomagnetic directional progression in Carboniferous rocks collected from immediately to the north of the Stellarton Sub-Basin across the Cobequid Fault, also suggest a revision in the timing of the reversal stratigraphy. In terms of the Stellarton Gap, this progression would be recorded by a small change in declination from

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Fig. 5. Vector diagrams demonstrating thermal demagnetization characteristics of samples from the Stellarton Group. Characteristic of samples from the Stellarton Group is the more moderately inclined vector. As shown by ST0611, many specimens exhibit random vector changes at temperatures above the Curie Point of magnetite.



around 180° to around 170° together with a gradual shallowing of the observed magnetic inclination from around 15° to 0°.

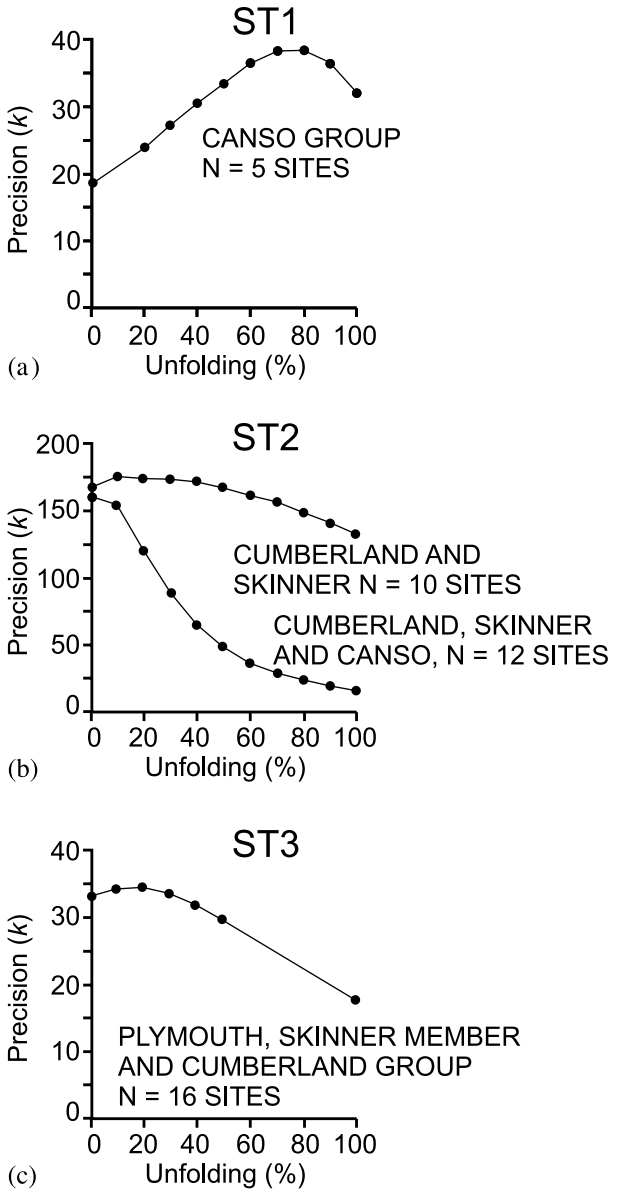
The declination of the ST1 is widely divergent from the anticipated direction. Either a progressive anti-clockwise rotation of the ST1 pole by 75°–85° or a clockwise rotation through 95°–105° can explain this divergence (Fig. 7B). Without appealing to arguments based on the polarity of the magnetization (the smaller clockwise rotation would imply the Canso Group at this locality has retained a reversed-polarity magnetization), one cannot differentiate which sense of rotation is applicable. Whichever is accepted, it should be clear that early strata within the Stellarton Graben have undergone considerable tectonic rotation (Fig. 7A). As the next youngest direction (ST2) does not exhibit such a widely divergent declination direction, most of this rotation must have occurred prior to the deposition of the Cumberland Group (Westphalian B to C).

While the declination data contains evidence for early lateral fault motions the inclination contains evidence for later vertical fault block motions. The rock units examined in this collection cover a very restricted time period of about 30 Ma from Namurian to Westphalian D. One would, therefore, not expect to see such large differences in remanence inclination as reported in Table 2. Further, the preferred sequence of remanence acquisition as outlined earlier in the text (ST1 – ST2 – ST3) does not conform to the commonly accepted pattern of a gradually shallowing inclination. ST3 with the shallowest inclination appears to be correctly placed as the youngest remanence in the sequence. But ST2, which has a steeper inclination than either ST1 or ST3, would superficially appear to be out of sequence. DiVenere and Opdyke (1991) observed exactly the same problem in their study of samples from the Joggins Section. The easiest explanation is, just like Divenere and Opdyke suggest, interpreting the ST2 component as the more ancient remanence even though, just like their B component, ST2 is characterized by lower temperature stability. In this case, the improvement of the Fisher precision upon 80% unfolding, together with the unconformity test of remanence age demands that the ST1 magnetization must predate the ST2 direction. When compared to the expected inclination, it would appear that the inclination of ST1 is too shallow, while the inclination of ST2 is much too steep. Pole ST3, as previously noted, conforms to the expected direction. Further evidence of complex vertical fault motions is contained in the mean ST2 grouping. As noted earlier in the text, the bedding attitudes recorded in the younger Skinner Member are steeper than in the underlying Cumberland Group.

Again, like DiVenere and Opdyke (1991), it is necessary to construct a tectonic model of remanence acquisition to explain these apparent discrepancies. Central to both the model

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Fig. 6. Plots of Fisher precision parameter (k) versus percentage unfolding for (a) the Canso Group, (b) the Cumberland Group, and (c) the Stellarton Group sites. Note: the peak for Canso Group is found with 80% unfolding, suggestive of syn-deformational remanence acquisition. Similar plots for ST2 and ST3 are suggestive of a post-deformational remanence acquisition.



presented here and that given by DiVenere and Opdyke (1991), is the concept that the local horizontal reference frame changes with time. In performing a simple fold test, it is generally assumed that it is correct to rotate each limb of the fold back to the *current* horizontal reference frame. If however, as might be expected in a pull-apart basin, differential vertical displacements occur on the two bounding faults, this will result in varying regional tilts in the basin with time. Hence the horizontal reference frame to which the magnetizations should be corrected may also be varying with time.

A model to resolve this inclination problem, while still conforming to the observed geological relationships, can be constructed as follows. Initially the Canso Group is deposited in the Stellarton Graben, which has been produced as a result of relative lateral motions between the Hollow and Cobequid faults. Later localized compression leads to folding of the Canso Group. The ST1 magnetization is acquired during this early phase of folding. Renewed dextral motions on the Hollow and Cobequid faults result in the strata within the graben being rotated by over 90° about a near-vertical axis. Continued vertical motion on the Hollow Fault produces a southerly dip in the Reidel Flake. The horizontal reference for the synfolding ST1 is now a surface dipping at ~15° to the south. There is both paleomagnetic and geological support for this interpretation. The more negatively inclined remanence direction detected at site 11 is probably a record of the original detrital remanent magnetization. The appropriate reference surface for the correction of this component then should be the present horizontal. Full-tilt correction of this vector would give an inclination of –10°. This closely agrees with the direction expected from the Canso Group and nicely explains the apparently shallow inclination of ST1. Geological evidence for significant downward vertical displacement on the Hollow Fault at this time is provided by the New Glasgow conglomerate, which was deposited as a northerly prograding alluvial fan complex (Yeo and Gao 1987).

The difference in tectonic dip between the Cumberland Group and Skinner Member can be explained by invoking a change in the relative amount of vertical motion on the two bounding faults. During and subsequent to the deposition of the Cumberland Group, downfaulting was greatest on the southern margin represented by the Hollow Fault. As a consequence, the Cumberland Group developed a shallow southerly dip. Renewed fault activity was signaled by the deposition of the Skinner Member of the Stellarton Group. During this phase of graben subsidence, the northerly boundary defined by the Cobequid Fault exhibited the greatest amount of downfaulting. This results in the Skinner Member acquiring a northerly dip. Since tectonic rotations are cumulative the observed dip of the Cumberland Group represents the vector sum of an initial southerly dip and the later northerly dip. The vector sum of these rotations results in an apparent shallower dip in the Cumberland Group. Acquisition of the ST2 remanence clearly postdates deformation of both the Cumberland Group and the Stellarton Group's Skinner Member and probably predates deposition of the Plymouth and Albion members. Since during the time represented by deposition of the Skinner and Plymouth members other studies indicate there was no significant change in remanence direction, it is clear that the observed inclination of ST2 must be too steep. This feature can be explained by further fault motion on the Cobequid Fault during deposition of the Plymouth and Albion members. Continued downfaulting on the northerly side of the Stellarton Sub-Basin would result in the reference frame for the ST2 direction acquiring a northerly dip. Correcting the ST2 directions to the present horizontal surface, instead of to a northerly dipping reference surface, would systematically increase the observed inclination. Using the dip observed in the Plymouth Member as an estimate for the amount of rotation that is applicable to this reference surface would correct the ST2 direction to a mean inclination of around +8°. The Coal

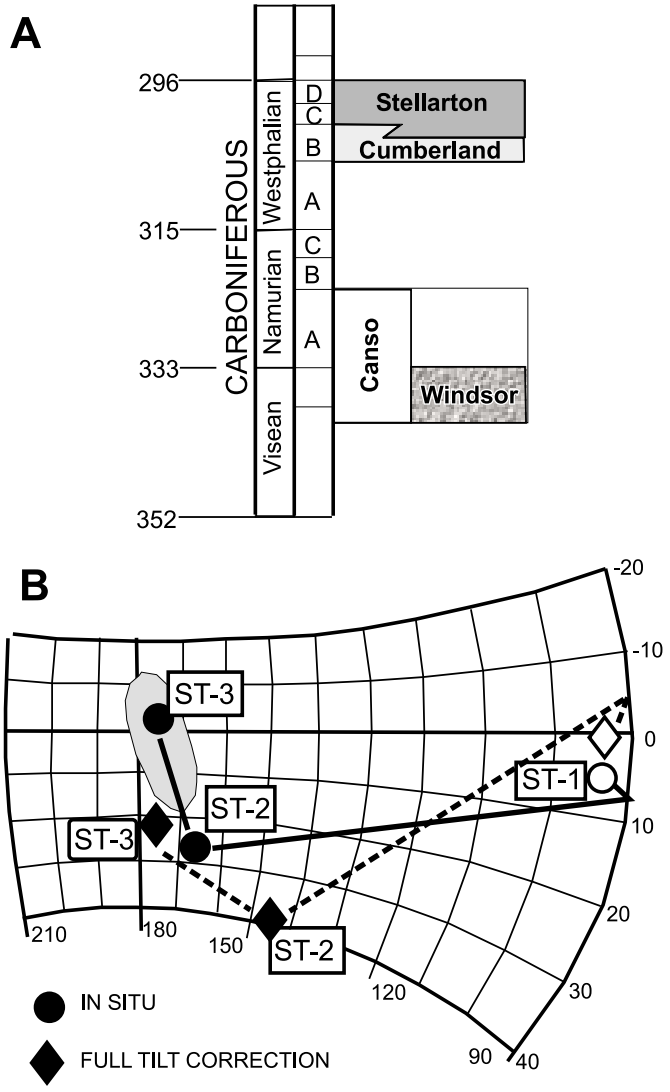
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Table 2. Mean remanence directions from carboniferous rocks of the Stellarton Basin.

Component	Correction	*N*	*R*	*k*	α_{95}	*D* (°)	*I* (°)	Long°E	Lat°N
ST3	0%	16	15.55	33.0	6.5	175.1	1.3	124.3	44.5
	20%	**16**	**15.56**	**34.3**	**6.4**	**174.9**	**5.6**	**124.3**	**41.4**
	100%	16	15.16	17.8	9.0	172.2	22.4	126.6	32.3
ST2	**0%**	**12**	**11.93**	**161.2**	**3.4**	**167.7**	**27.7**	**131.1**	**28.7**
	100%	12	11.30	15.7	11.3	155.7	44.4	139.9	14.9
ST1	0%	5	4.78	18.6	18.2	81.4	5.4	31.8	7.9
	80%	**5**	**4.90**	**38.4**	**12.5**	**83.9**	**1.5**	**31.3**	**4.8**
	100%	5	4.88	32.2	13.7	83.9	0.8	31.6	4.6

Note: Results from the Forbes Lake Conglomerate are not included in any mean vector calculation.

Fig. 7. (A) Stratigraphic section for Stellarton Gap showing approximate age of each of the units sampled. (B) Stereonet plot (equatorial projection viewed from south) showing change of remanence direction during the Carboniferous for the Stellarton Graben. Expected path for area as defined by Van der Voo (1993) is marked by stippled zone. It is argued that magnetizations in Stellarton Graben were acquired in the sequence ST1 – ST2 – ST3.



Brook and Thorburn members record the final phase of subsidence in the Stellarton graben. The extent of displacement of the Plymouth and Albion reference surface during this last period of subsidence is not defined due to the lack of any younger remanence data.

Regional tectonic implications

Many of the sub-basins of the Carboniferous Maritimes Basin occur in close proximity to major tectonic lineaments (Fig. 1). Morris (1976) suggested an observed disparity between Paleozoic paleomagnetic results from Britain and North America could be explained by large-scale transcurrent displacement within the Appalachian Orogen during the Paleozoic. Later, Kent and Opdyke (1979) reported that upper Paleozoic results from sedimentary basins located within the orogenic belt gave paleomagnetic poles that were divergent to those from cratonic locales. Again the discrepancy was interpreted as evidence for large-scale (>1500 km) lateral translation of a large sliver (Acadia) of the orogenic belt (Kent and Opdyke 1979). This interpretation was questioned when Irving and Strong (1984) reported a pole from Carboniferous sediments of the Deer Lake basin that was not statistically distinct from other poles from within the orogenic belt. Irving and Strong then argued that since the sediments of the Deer Lake basin are sitting on Grenville-age basement rocks any large-scale lateral displacement was not a viable explanation. By questioning the validity of the cratonic poles, Irving and Strong (1984) considered the discrepancy identified by Kent and Opdyke (1979) to be more apparent than real, suggesting that the results could be accommodated by "a systematic time variation of paleolatitude". Subsequent studies (Johnson and Van der Voo 1989) and this new study have confirmed Kent and Opdyke's (1979) original observation, paleomagnetic poles from locations with the orogenic belt are distinct from those sampled on the craton (Fig. 8).

As demonstrated by Irving et al. (1980) displacement of poles from orogenic belts relative to their expected position on a cratonic reference polar path can be explained by either lateral displacement or tilting. While Irving and Strong (1984) argued that the presence of Grenville basement under the Dear Lake sediments precluded any large-scale lateral translation, this statement was based on the unproven assumption of no displacement between the Grenville basement at Dear Lake and the main Grenville orogenic belt. The qualitative argument was based solely on the presently

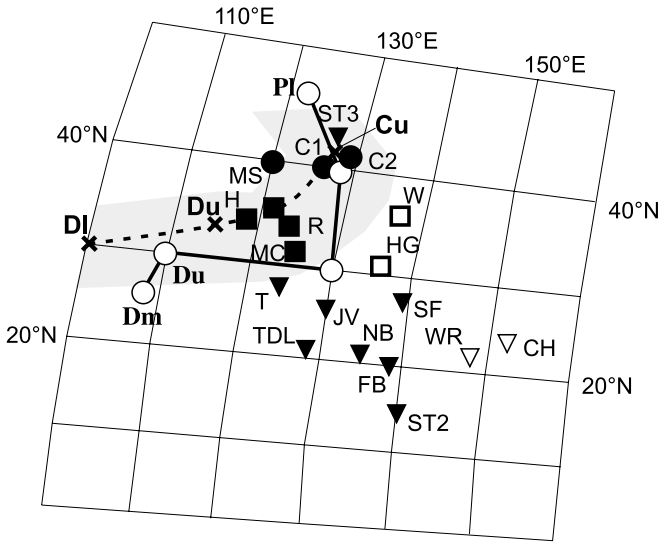
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observed distribution of Grenville-age rocks. The available evidence does not preclude the possibility of some lateral displacement. Many of the studies that give poles in the orogenic group are based on magnetizations that were demonstrably acquired penecontemporaneously with local folding events (Johnson and Van der Voo 1999), just like the ST2 magnetization in this study. In this situation, as noted by DiVenere and Opdyke (1991), the importance of defining exactly to which horizontal reference frame rotations are calculated becomes critical. This becomes essential when, as described by geological studies, the polarity of the half-grabens controlling sedimentation is changing both temporally and spatially (Hamblin and Rust 1989). It is very likely then that the observed discrepancy arises from unrecognized tilt acquired subsequent to remanence acquisition.

Pole ST1 is widely divergent from most other Carboniferous poles. Studies by Tanczyk (1992) on rock units from the Meguma Terrane have also yielded widely divergent pole positions. Such disparate directions are also known from other sectors of the Appalachian–Caledonide orogenic belt. Piper (1991) for example has reported evidence for rotation of tectonic slivers about vertical axes in the Caledonides of western Ireland. Like the Maritimes basins discussed earlier in the text, the upper age limit for these rotations appears to be mid-Carboniferous.

Pole ST2 is in agreement with a number of other poles from within the orogenic belt of the Maritimes Basin (Fig. 8). These poles, however, are displaced from the reference pole path of Van der Voo (1993) and composed of both cratonic and orogenic poles and a purely cratonic path based on Roy and Morris (1983). This discrepancy can be explained either by rotation around a vertical axis, or incorrect definition of the local horizontal (McDonald 1981). The spatial geometry of the study site and pole position requires that rotation about a vertical axis would result in the poles being displaced to the southeast of the correlative cratonic poles. Hence some element of rotation about a horizontal axis is necessary. With the current information, there are a number of possible combinations of vertical rotation and tilt that can provide the appropriate rotation. Pole ST3 is in close agreement with a significant number of poles from other Late Carboniferous rock units (Fig. 8).

Fig. 8. Polar plot based on original fig. 9 of Johnson and Van der Voo (1989). Pole path for cratonic North America, as defined by dotted line and stippled area, is based on curve published by Roy and Morris (1983). Average pole path for North America represented by solid line is based on Van der Voo (1993). Early Carboniferous poles are represented by triangles, Mid-Carboniferous poles are represented by squares, and Late Carboniferous results are represented by circles. Open symbols represent sites from the Meguma Terrane. ST1 does not plot on this diagram. ST2 plots with other displaced sites from the Maritimes sedimentary basins. ST3 is identical to common overprint direction. Other poles: CM, Combined pole for Late Carboniferous of Maritimes (Johnson and Van der Voo (1989); CC, Combined pole for Late Carboniferous of craton (Johnson and Van der Voo (1989); H, Hopewell (Roy and Park 1969); R, Riversdale; MS, Maringouin and Shepody (Roy and Park 1974); W, Windsor (Scotese et al. 1984); HG, Horton (Scotese et al. 1984); MC, Mauch Chunk (Kent and Opdyke 1985); JV, Jeffreys Village (Murthy 1985); DL, Dear Lake (Irving and Strong 1984); T, Terenceville (Kent 1982); SF, Spout Falls (Murthy 1985); FB, Fisset Brook (Johnson and Van der Voo 1989); NB, New Brunswick (Seguin et al. 1985); CH, Chevrie (Spariosu et al. 1984); WR, White Rock (Spariosu et al. 1984).



Conclusions

Most of the sub-basins of the Carboniferous Maritimes Basin are separated from each other by major tectonic discontinuities. The timing, sense, and lateral extent of motions on these faults have been the subject of extensive discussion in the literature (see for example the reviews in St. Julien and Beland (1982) and Beaumont and Tankard (1987)). In question is the amount of lateral versus vertical motion on these faults. The two extremes are the rift model proposed by Belt (1968), in which vertical motions predominate, and the strike-slip pull-apart model proposed by Bradley (1982) in which lateral motions predominate. Boehner et al. (1986), followed by Ryan et al. (1987) proposed a compromise in which they interpret the Basin as having developed in two distinct phases. The first stage, extending from the Upper Devonian to mid-Westphalian, is identified as the pull-apart phase and is related to regional tectonic transpression. The second phase in the development of the basin, from late Westphalian to Early Permian, is considered as the thermal subsidence phase. Paleomagnetically, it should be possible to differentiate between these two models because they produce quite different tectonic signatures. Christie-Blick and Biddle (1985) show that progressive development of a pull-apart basin produces incrementally increasing rotations about vertical axes in sequentially older strata deposited. The sense of rotation of these blocks ("flakes") is directly related to the sense of motion on the basin-bounding faults. In contrast development of a purely subsidence generated basin involves only broad-scale regional tilting. More recently, Calder (1998) in a comprehensive review of Maritime Carboniferous basins has confirmed the importance of the two-phase model of basin development.

Most of the earlier paleomagnetic results from the Maritimes basin reported by Roy and his co-workers (Roy 1963; Roy and Park 1969, 1977, and others) were from collections in the Moncton sub-basin and from the western end of the

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Cumberland Basin. From these results Roy and Morris (1983) produced a Maritimes Basin Carboniferous reference curve that defined a simple systematic change of remanence direction with increasing age. Later publications, for example Irving and Strong (1984) and Johnson and Van der Voo (1989), reported poles from other rock units in other parts of the Maritimes Basin that did not fall on the cratonic curve of Roy and Morris (1983). Data from Carboniferous rocks overlying the most outboard suspect terrane, the Meguma zone, have been shown to be systematically divergent from other Carboniferous data (Spariosu et al. 1984; Scotese et al. 1984, and Fig. 8). All these divergent results are suggestive of tectonic rotation of slivers within the orogenic belt relative to the craton.

Paleomagnetic studies of sequential stratigraphic units from the Stellarton Sub-Basin, which is located south of the Cobequid Fault but north of the Hollow Fault, define two periods of basin development. Pole ST1, which is found only in the oldest unit sampled, the Canso Group, indicates this unit has been rotated by at least 90°. Such clockwise rotation is consistent with the sense of motion that one might expect of a "Riedel flake" located between two dextral strike-slip faults (Dewey 1982; Christie-Blick and Biddle 1985). And as shown by Yeo and Gao (1987), both the Eastern Cobequid and Hollow faults exhibit evidence for Late Carboniferous dextral strike-slip motion. Assuming a maximum width for the graben of 6 km, simple calculations suggest that the maximum total lateral displacement on the bounding faults to be 7–20 km. The next youngest remanence ST2 does not exhibit a widely divergent declination. Therefore, most of the fault translation must have occurred prior to the deposition of the Cumberland Group (Westphalian B to C).

To interpret the inclination data in terms of the preferred sequence of remanence acquisition (ST1 – ST2 – ST3), it is necessary to invoke the idea of sequential changes in the regional tilt of the basin and hence the local horizontal reference frame. DiVenere and Opdyke (1991) used the same approach to explain a similar apparent contradiction. Such changes in regional attitude are to be expected when there are differential vertical motions on the basin-bounding faults. Geological and paleomagnetic evidence suggests that during the deposition of the Canso and Cumberland groups displacement was greatest on the southern margin of the Stellarton Sub-Basin, the Hollow Fault. The result of this is that the older strata developed an initial southerly dip. During deposition of the Stellarton Group, the northerly margin of the Basin, the Cobequid Fault, shows the greatest amount of displacement. The younger strata in the Stellarton Sub-Basin now have a moderate northerly dip. The discrepancy exhibited by both the ST2 pole reported here and a number of other poles from basins within the orogenic belt suggests this second stage tilting may have been a common feature of Carboniferous Maritimes Basin.

Paleomagnetic results from the Stellarton Graben define two distinctly different periods of basin development. Prior to Westphalian B to C time, basin formation was related to strike-slip transpression produced by dextral motion on the eastern Cobequid and Hollow faults. The Stellarton Graben formed as a simple rhomb graben caused by regional northwest–southeast compression. Subsequent to Westphalian B to C time, further development of the Stellarton Graben was caused by mainly vertical movements on the basin-bounding faults. With time, the maximum fault displacement moved from the Hollow Fault to the Cobequid Fault. These paleomagnetic observations confirm the two-phase model of Maritime Basin genesis as proposed by Bradley (1982), Boehner et al. (1986), Ryan et al. (1987), and Calder (1998) on the basis of geological mapping.

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I would like to dedicate this paper to the memory of an old friend, Jean Roy. While a Post-Doctorate Fellow, at the then Earth Physics Branch in Ottawa, Jean introduced me to the subtleties of remanence acquisition in red beds. Careful and insightful reviews by David Symons, Ken Buchan and Fred Cook, the Associate Editor, have resulted in a vastly improved manuscript. None of this work would have been possible without the guidance of Gary Yeo; thanks. Initial funding for this project was provided through a contract to Morris Magnetics Inc. from a Canada – Nova Scotia Mineral Development Agreement 1984–1989.

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